# EXISTENCE OF THE SECOND ADAMS OPERATION ON K_0 OF THE VECTOR BUNDLES OF A SCHEME, WITH ADDITIVITY, PULLBACK COMPATIBILITY AND MULTIPLICATIVITY

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

*Editorial note.* The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained apart from the notation of K_0(Vect(X)) which the statement uses without recalling it.

*Editorial note.* Canonical location of the counted proposition: Stacks tag 0FEJ (sources/stacks-chow/tags.txt line 19254).

*Allowed standard prerequisites (editorial).*

- The Grothendieck group K_0 of finite locally free modules and its presentation by short exact sequences. Algebra, Section 'K-groups' (tag 00JC region) and the same chapter's earlier K_0 material; the statement speaks of K_0(Vect(X)) without recalling it. Supplies that K_0(Vect(X)) is generated by classes of finite locally free modules subject to the relations coming from short exact sequences, which is the relation used at proof lines 12797-12800 to show that the formula psi^2 = [Sym^2] - [wedge^2] descends to a well defined additive map.

**Lemma 0.1.** *Let $X$ be a scheme. There is a ring map*

$$\psi^2 : K_0(\mathit{Vect}(X)) \longrightarrow K_0(\mathit{Vect}(X))$$

*which sends $[\mathcal{L}]$ to $[\mathcal{L}^{\otimes 2}]$ when $\mathcal{L}$ is invertible and is compatible with pullbacks.*

*Proof.* Let $X$ be a scheme. Let $\mathcal{E}$ be a finite locally free $\mathcal{O}_X$-module. We will consider the element

$$\psi^2(\mathcal{E}) = [\mathrm{Sym}^2(\mathcal{E})] - [\wedge^2(\mathcal{E})]$$

of $K_0(\mathit{Vect}(X))$.

Let $X$ be a scheme and consider a short exact sequence

$$0 \to \mathcal{E} \to \mathcal{F} \to \mathcal{G} \to 0$$

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of finite locally free $\mathcal{O}_X$-modules. Let us think of this as a filtration on $\mathcal{F}$ with 2 steps. The induced filtration on $\mathrm{Sym}^2(\mathcal{F})$ has 3 steps with graded pieces $\mathrm{Sym}^2(\mathcal{E})$, $\mathcal{E} \otimes \mathcal{F}$, and $\mathrm{Sym}^2(\mathcal{G})$. Hence

$$[\mathrm{Sym}^2(\mathcal{F})] = [\mathrm{Sym}^2(\mathcal{E})] + [\mathcal{E} \otimes \mathcal{F}] + [\mathrm{Sym}^2(\mathcal{G})]$$

In exactly the same manner one shows that

$$[\wedge^2(\mathcal{F})] = [\wedge^2(\mathcal{E})] + [\mathcal{E} \otimes \mathcal{F}] + [\wedge^2(\mathcal{G})]$$

Thus we see that $\psi^2(\mathcal{F}) = \psi^2(\mathcal{E}) + \psi^2(\mathcal{G})$. We conclude that we obtain a well defined additive map $\psi^2 : K_0(\mathit{Vect}(X)) \to K_0(\mathit{Vect}(X))$.

It is clear that this map commutes with pullbacks.

We still have to show that $\psi^2$ is a ring map. Let $X$ be a scheme and let $\mathcal{E}$ and $\mathcal{F}$ be finite locally free $\mathcal{O}_X$-modules. Observe that there is a short exact sequence

$$0 \to \wedge^2(\mathcal{E}) \otimes \wedge^2(\mathcal{F}) \to \mathrm{Sym}^2(\mathcal{E} \otimes \mathcal{F}) \to \mathrm{Sym}^2(\mathcal{E}) \otimes \mathrm{Sym}^2(\mathcal{F}) \to 0$$

where the first map sends $(e \wedge e') \otimes (f \wedge f')$ to $(e \otimes f)(e' \otimes f') - (e' \otimes f)(e \otimes f')$ and the second map sends $(e \otimes f)(e' \otimes f')$ to $ee' \otimes ff'$. Similarly, there is a short exact sequence

$$0 \to \mathrm{Sym}^2(\mathcal{E}) \otimes \wedge^2(\mathcal{F}) \to \wedge^2(\mathcal{E} \otimes \mathcal{F}) \to \wedge^2(\mathcal{E}) \otimes \mathrm{Sym}^2(\mathcal{F}) \to 0$$

where the first map sends $ee' \otimes f \wedge f'$ to $(e \otimes f) \wedge (e' \otimes f') + (e' \otimes f) \wedge (e \otimes f')$ and the second map sends $(e \otimes f) \wedge (e' \otimes f')$ to $(e \wedge e') \otimes (ff')$. As above this proves the map $\psi^2$ is multiplicative. Since it is clear that $\psi^2(1) = 1$ this concludes the proof. $\square$